

A woman with blonde hair, wearing a brown dress, sits on a grassy hill and plays an acoustic guitar. She is surrounded by a group of children, mostly girls, who are sitting on the grass and looking towards her. The background features a vast, green mountain landscape with rolling hills and a small village in the distance. The scene is bathed in soft, natural light, suggesting a peaceful outdoor setting.

**LET'S START AT THE VERY
BEGINNING**

Ch.1 Introduction

**IT'S A VERY GOOD PLACE TO
START**

Practical Applications of this book/course

- Improving self-control
- Understanding eating disorders
- Overcoming fears and probias
- Improving relationships with others
- Raising children

Study Tips

- Read a short section of material
- Recite the material to yourself, without looking at the material
 - What do you remember and not remember
- Review the material again
- The perils of highlighting
- Effortful
- The perils of internet advice

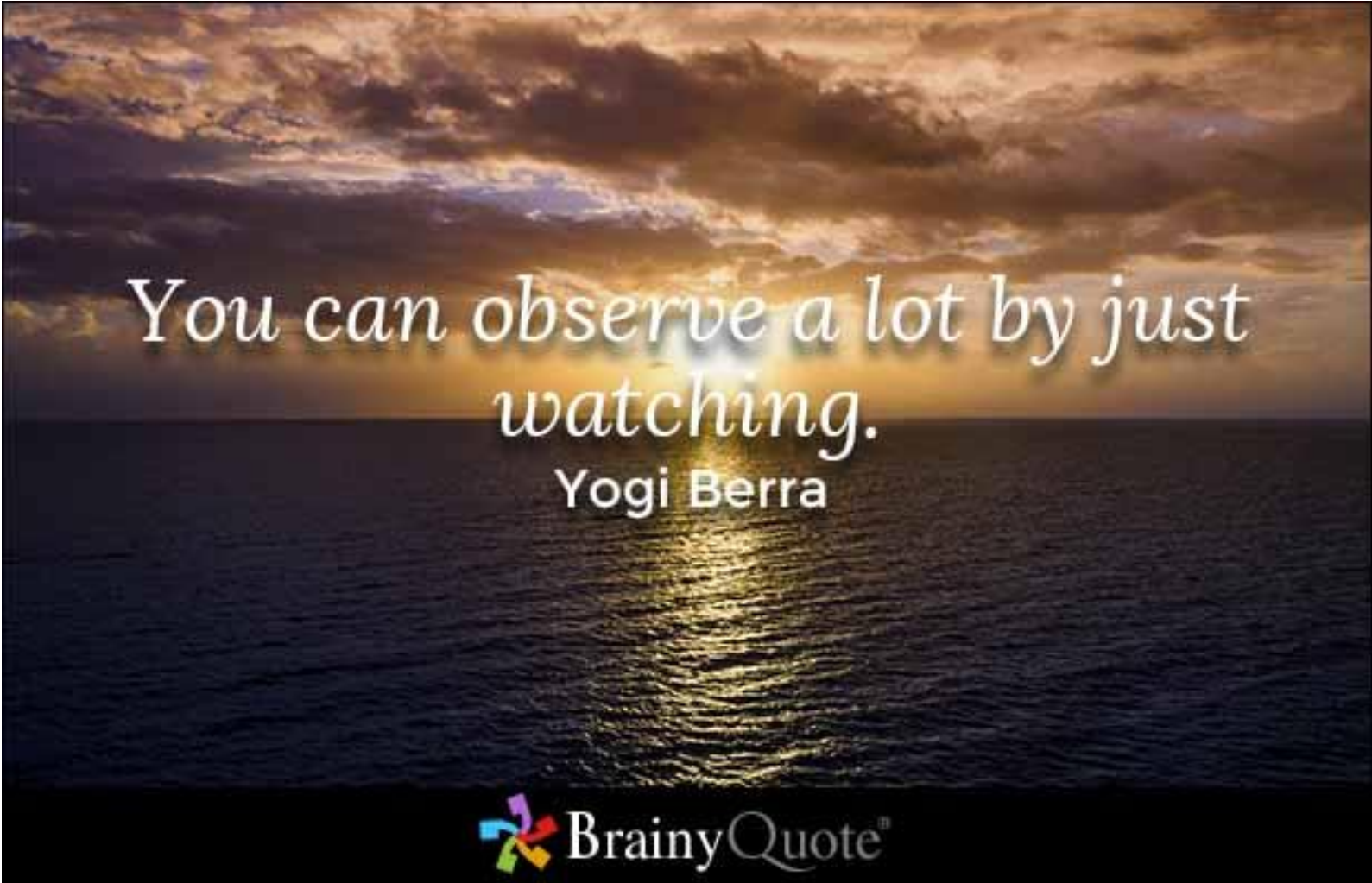


What is one of the best ways to learn the material?



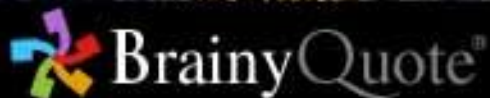
What is learning and behavior?

- Behavior
 - Any activity of an organism that can be observed or somehow measured
- Learning
 - Relatively permanent change in behavior that results from some type of experience



*You can observe a lot by just
watching.*

Yogi Berra



AH YES, THE PAST CAN HURT...

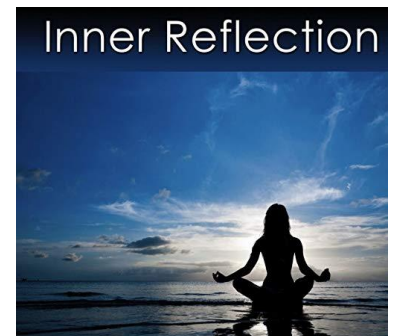


**BUT THE WAY I SEE IT YOU CAN EITHER
RUN FROM IT, OR LEARN FROM IT**

HISTORICAL BACKGROUND

Nativist (Nature)

- Assumes that a person's abilities and behavioral tendencies are largely inborn.
- Plato believed that everything we know is inborn.
- Learning is simply a process of inner reflection to uncover the knowledge that already exists within.



Empiricist (Nurture)

- Assumes that a person's abilities and tendencies are mostly learned.
- Aristotle disagreed with Plato.
- He argued that knowledge is not inborn but instead is acquired through experience.



4 Laws of Association

1. Law of Similarity – events/objects similar to one another are readily associated with each other



4 Laws of Association

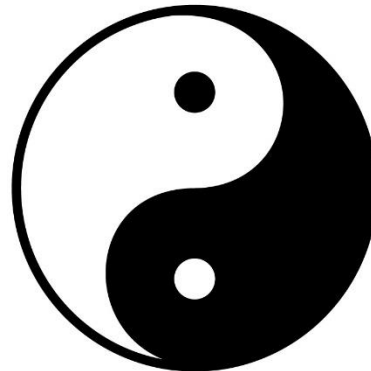
2. Law of Contrast – events/objects opposite from each other are readily associated

Order and chaos constitute the stage

Known

Yang

Order
Masculinity
Day
The known
Authoritarianism
Fascism



Unknown

Yin

Chaos
Femininity
Night
The unknown
Decadence
Nihilism

4 Laws of Association

3. Law of Contiguity – events that occur in close proximity to each other are readily associated



4 Laws of Association

4. Law of Frequency – the more frequent 2 items occur together the more strongly they are associated



René Descartes



- Mind-body dualism
 - Some behaviors are reflexive and “machine-like”
 - Some behaviors are under control of “free will”
 - Only humans have “free will”
- The reflexive behaviors can be studied scientifically, even in animals, to better understand the human condition

British Empiricist



- John Locke
 - All knowledge is a function of experience
 - Tabula Rasa: blank slate
- Conscious mind is composed of a combination of a finite set of basic elements
 - Association creates complex sensation and thought patterns



Structuralism

- Holds that it is possible to determine the structure of the mind by identifying the basic elements that compose it.
- The method of *introspection* is when the subject in an experiment attempts to accurately describe his or her conscious thoughts, emotions, and sensory experiences.
- The emphasis on systematic observation helped establish psychology as a scientific discipline.

Functionalism

- Assumes that the mind evolved to help us adapt to the world around us and that the focus of psychology should be the study of those adaptive processes.
- Psychologists should study the adaptive significance of the mind.
- E. B. Thorndike and John B. Watson, two of the most important figures in behaviorism, were students of functionalist psychologists.

Theory of Evolution

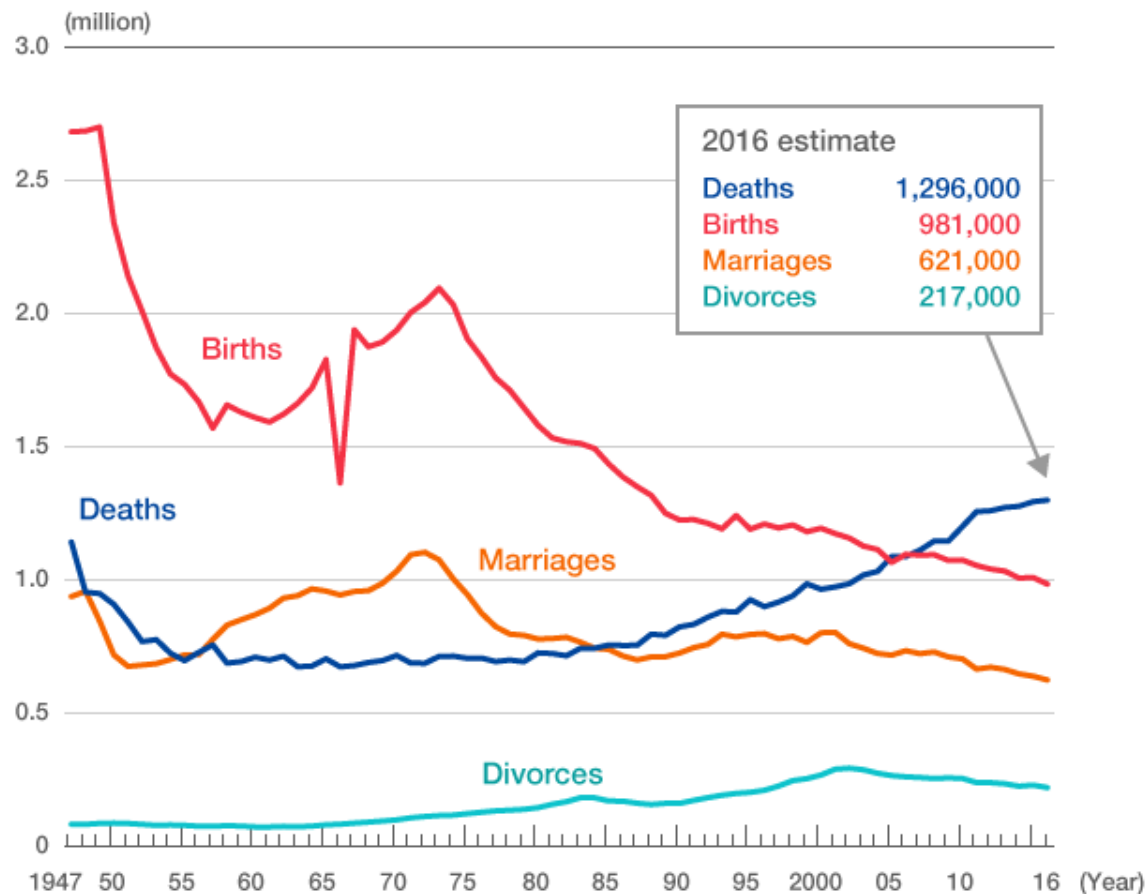
- *Natural selection* is the concept that individuals or species that are capable of adapting to environmental pressures are more likely to survive and reproduce than those that cannot adapt.
- 3 Components
 - *Traits vary,*
 - *many traits are heritable, and*
 - *organisms must compete for limited resources.*

The real driving force behind evolution is not survival of the fittest...

but the *reproductive advantage that accrues to those individuals possessing traits that are best suited to the environment.*

Japan's Annual Birth Rate

Births/Deaths and Marriages/Divorces by Year



Note: Calculations based on the estimated total population as of October 1, 2016 (125,245,000 people)

Source: Ministry of Health, Labor, and Welfare demographic survey

Sperm Count in Western Men Has Dropped Over 50 Percent Since 1973, Paper Finds

By MAYA SALAM AUG. 16, 2017



Total sperm count among Western men fell 59.3 percent from 1973 to 2011, according to new research.
John Walsh/Photo Researchers

The sperm count of men in Western countries has been declining precipitously with no signs of “leveling off,” [according to new research](#), bolstering a school of thought that male health in the modern world is at risk, possibly threatening fertility.

By examining thousands of studies and conducting a meta-analysis of 185 — the most comprehensive effort to date — an international team of researchers

RELATED COVERAGE



Opinion | Op-Ed Contributor
How to Save Your Sperm JULY 31, 2017



Sperm Counts via Your Cellphone
MARCH 22, 2017



Opinion | Nicholas Kristof
Are Your Sperm in Trouble? MARCH 11, 2017

Semen samples from 42,935 men from 50 countries from 1973 to 2011. the number of sperm per milliliter of semen — had declined each year, amounting to a 52.4 percent total decline, in men from North America, Europe, Australia and New Zealand.

Why?



**SMOKING MARIJUANA COULD
LOWER SPERM COUNT**

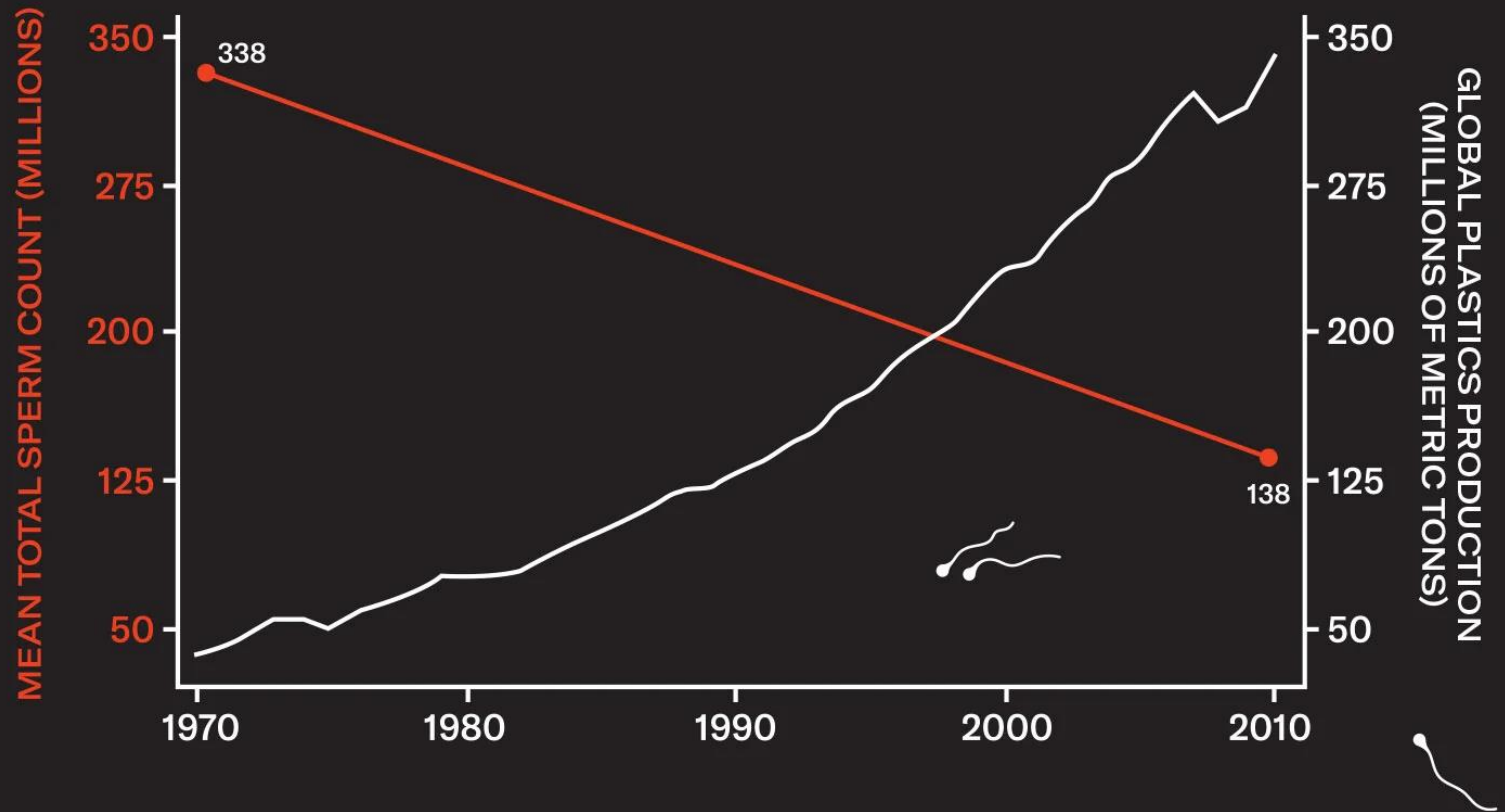


TOP 10 FOODS TO INCREASE SPERM COUNT

Increase Fertility In Men
100% Natural Ways



MEAN SPERM COUNT AND GLOBAL PLASTICS PRODUCTION SINCE 1970



99% of these sea turtles were born girls. Is global warming to blame?

USA TODAY NETWORK Sean Rossman, USA TODAY Published 3:54 p.m. ET Jan. 9, 2018 | Updated 6:13 p.m. ET Jan. 10, 2018



(Photo: Andrey Danilovich, Getty Images)

CONNECT TWEEET LINKEDIN 1 COMMENT 3 EMAIL MORE

Researchers suggest a generation of sea turtles in the Great Barrier Reef were born mostly female because they were nested in warmer areas, raising concerns global warming might threaten the species.

The study, published Monday in the journal *Current Biology*, found 99.8% of green sea turtles near adulthood and originating from the northern — and warmer — part of the Great Barrier Reef were born female. A slightly younger group of juvenile turtles was found to be 99.1% female.

"It was surprising and we were not expecting it at all," said co-author Camryn Allen of the Joint Institute for Marine and Atmospheric Research in Honolulu, Hawaii. "We didn't expect it to be that extreme."

The study analyzed more than 400 turtles and was conducted by researchers from the U.S. National Oceanic and Atmospheric Administration (NOAA), as well as the Queensland Department of Environment and Heritage Protection in Australia.

The lopsided gender split, the authors point out, could cause the population to collapse, or the species to go extinct, unless efforts are made to lower nesting temperatures.

Read more: [395 cold-stunned sea turtles released back into Gulf of Mexico](#)

In the southern part of the Great Barrier Reef, where sand temperatures are cooler, females still outpaced males, but by far less. The study found juvenile and subadult turtles were 67.8% and 64.5% female, respectively.

Business Travelers



LA QUINTA INN



Former general says he knows how powerful North Korea's military is
www.usatoday.com



Missouri Gov. Eric Greitens admits to affair, denies blackmail
www.usatoday.com



Walmart boosts minimum wage to \$11, hands out bonuses up to \$1,000 for
www.usatoday.com

Parthenogenesis. It's extremely rare among vertebrates: Only 70 backboned species can do it, which is about 0.1% of all vertebrates

https://www.cnn.com/2020/03/09/us/komodo-dragons-parthenogenesis-scen-trnd/index.html

CNN travel Destinations Food and Drink Stay News Videos

LIVE TV Edition

A Komodo dragon with no male partner gave birth to three hatchlings

By Alicia Lee, CNN
Updated 12:44 AM ET, Tue March 10, 2020



Female Komodo dragon gives birth without male partner

Netflix releases 'Enola Holmes' trailer featuring Millie Bobby Brown

See four-year-old hit a hole-in-one

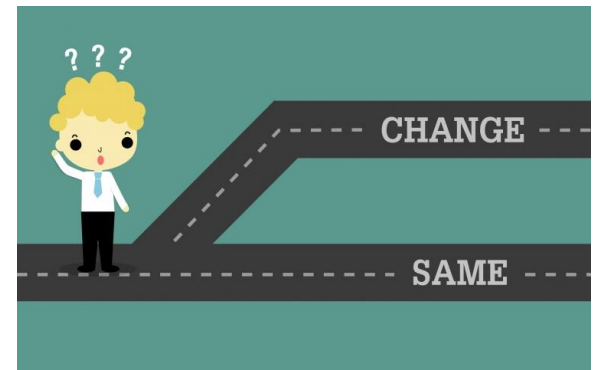
Kevin Hart calls NBC's Usain Bolt mistake 'disrespectful'

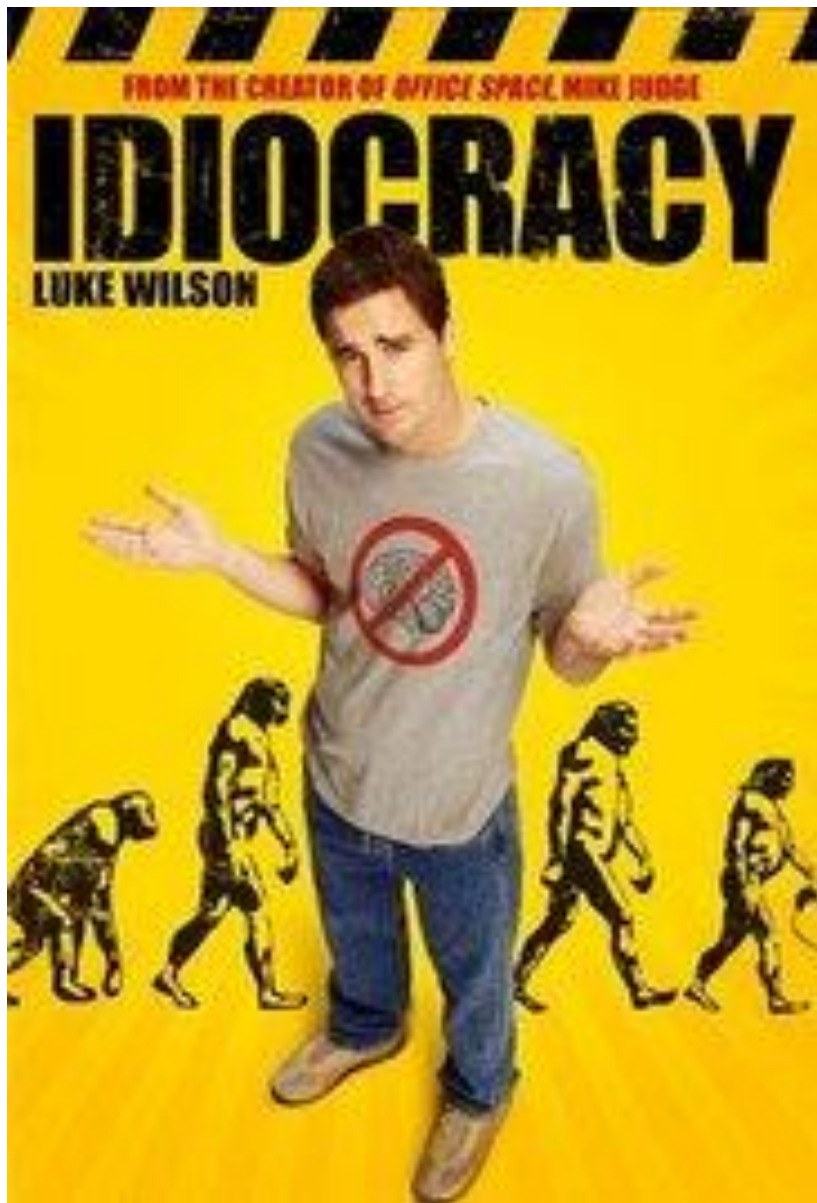
See what's Jennifer G: slow motion

(CNN) — Charlie, a female Komodo dragon at the Chattanooga Zoo in Tennessee, has proved to be the ultimate independent lady after successfully giving birth to three hatchlings without a male partner.

Evolutionary Adaptation

- A trait that evolves as a result of natural selection
- Individuals with successful traits are more likely to have offspring who, when they inherit the successful traits from their parents, are also more likely to survive and reproduce.
- These traits can be both physical adaptations & behaviors.





The Ability to Learn

- Evolved because it conferred significant survival advantages to those who had this ability.



Behaviorism

- The study of environmental influences on observable behavior.
- We are unable to directly observe another person's thoughts and feelings.
- James Watson reasoned that for psychology to be a purely “objective science”, it must be based solely on the study of directly observable behavior and the environmental events that surround it.
 - Watson's goal was for there to be a natural science approach to psychology, thus far a largely philosophical field

A possible introspection argument

- If you fail to reproduce my findings, it is not due to some fault in your apparatus or in the control of your stimulus, but it is due to the fact that your introspection is untrained...If you can't observe 3-9 states of clearness in attention, your introspection is poor. If, on the other hand a feeling seems reasonably clear to you, your introspection is again faulty. You are seeing too much. Feelings are never clear (Watson, 1913, p 163)

Measurement

- Objective
- Verifiable



Law of Parsimony

- simpler explanations for a phenomenon are generally preferable to more complex explanations.
- *Morgan's Canon* argued that we should interpret an animal's behavior in terms of
 - lower, more primitive processes (e.g., reflex or habit)
 - rather than higher, more mentalistic processes (e.g., decision or imagination).
- Watson thought the same should be true for interpretations of human behavior.

The Wisdom of
NAHUM
for the New Millenium

1. Never get discouraged.

2. Get a good protocol
that works fast.

3. Always make
the phone
calls in the
morning.

4. Never kill
the antibody
producing
rabbit.

5. Data have to be
nice, preferably
superb.

6. Trust NO one: You
have to see it with
your own eyes.

7. When repeating an
unsuccessful experiment,
make only one
change.

8. How do you
know that what
is in the tube is
what you think
is in the tube?

9. Ask simple
questions to
obtain simple
answers.

10. You can't be
first and wrong
or second and
right. Always
be first and right.

**SONENBERG
LABORATORY**

Five Schools of Behaviorism

Watson's Methodological Behaviorism

- Psychologists should study only publicly observable behavior.
- Events that can only be
 - subjectively perceived (such as our inner thoughts and feelings)
 - or that are assumed to exist on an unconscious level (such as a mother's unconscious hatred of her unwanted child)were to be stricken from scientific analysis.
- Few behaviorists were this extreme.

Methodological Behaviorism

FIGURE

perceive
from the
changes

sciously
cluded
ween

MENTALISM

**WATSON'S
BEHAVIORISM**

Env

ble
or

Dr. John Watson

- **BIOLOGICAL DETERMINISM**

- **TABULA RASA**

THE BELL CURVE

Intelligence and Class Structure
in American Life



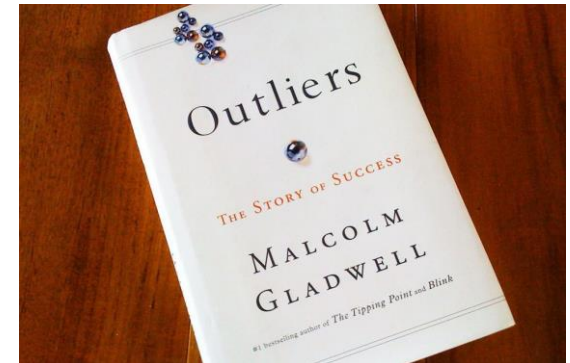
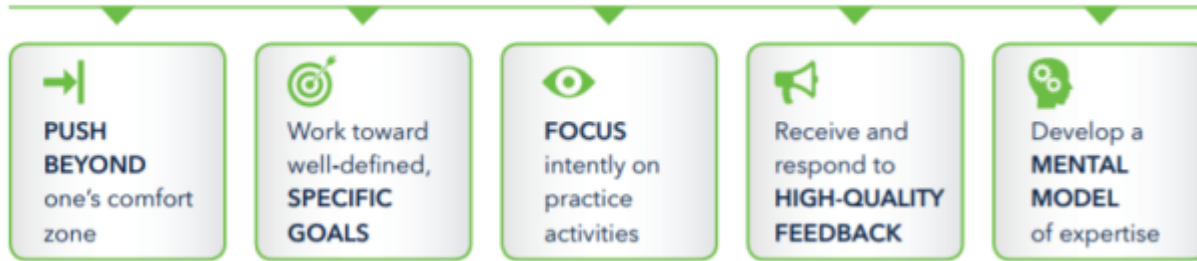
RICHARD J. HERRNSTEIN
CHARLES MURRAY

What is Growth Mindset?



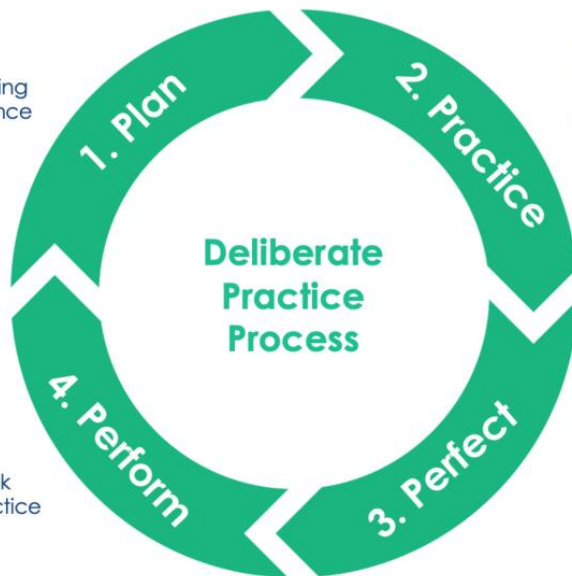
Deliberate Practice

THE FIVE PRINCIPLES OF DELIBERATE PRACTICE



PLAN

- Set goals
- Basic understanding of high performance
- Identify potential barriers
- Recruit coach
- Design practice



PRACTICE

- Practice deliberately
- Maintain motivation
- Overcome barriers + plateaus
- Work with coach

DELIBERATE PRACTICE



PERFORM

- Perform
- Self-evaluate
- Receive feedback
- Inform future practice
- Evolve plan
- Assess goals

PERFECT

- Self-evaluate
- Receive coach feedback
- Evolve practice
- Develop clearer mental object

Deliberate Practice

- Does not mean there is no role for hereditary input to expertise



Hull's Neobehaviorism

- Hull claimed that Watson's rejection of unobservable events was scientifically unsound.
- Hull noted that both physicists and chemists make inferences about events that have never been directly observed but that can nevertheless be *operationalized*.

Dark matter: https://youtu.be/QAa2O_8wBUQ

- 96% of the universe is unknown
- 4% consists of stars, planets, and galaxies
- Rest is dark energy and dark matter

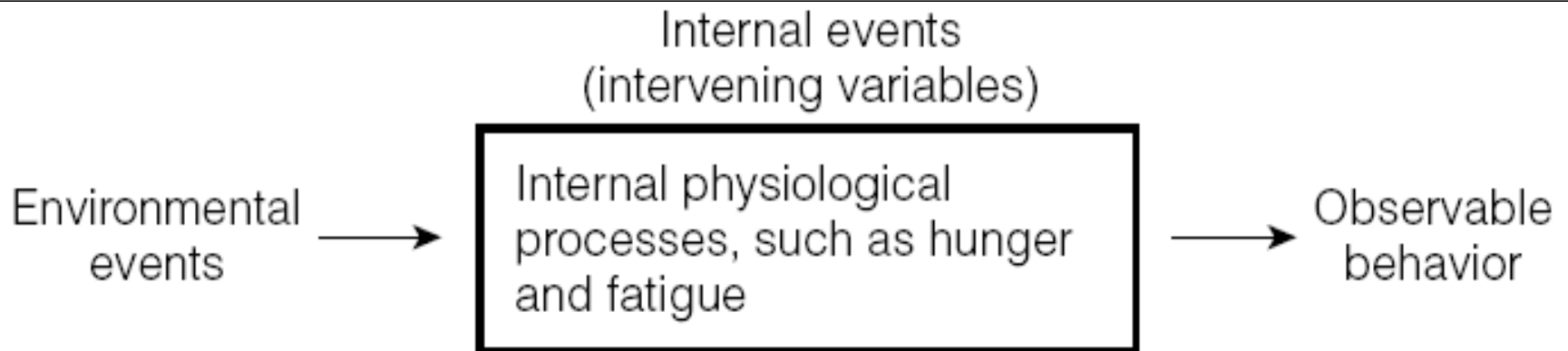
Hull's Neobehaviorism

- Psychologists should infer the existence of internal events that might *mediate* between the environment and behavior.
- Mediators largely consisted of physiological-type reactions.
- Example: a “hunger drive” that can be operationalized as number of hours of food deprivation.

Neobehaviorism

FIGURE 1.2

In Hull's neobehaviorism, theorists make use of intervening variables, in the form of hypothesized physiological processes, to help explain the relationship between the environment and behavior.



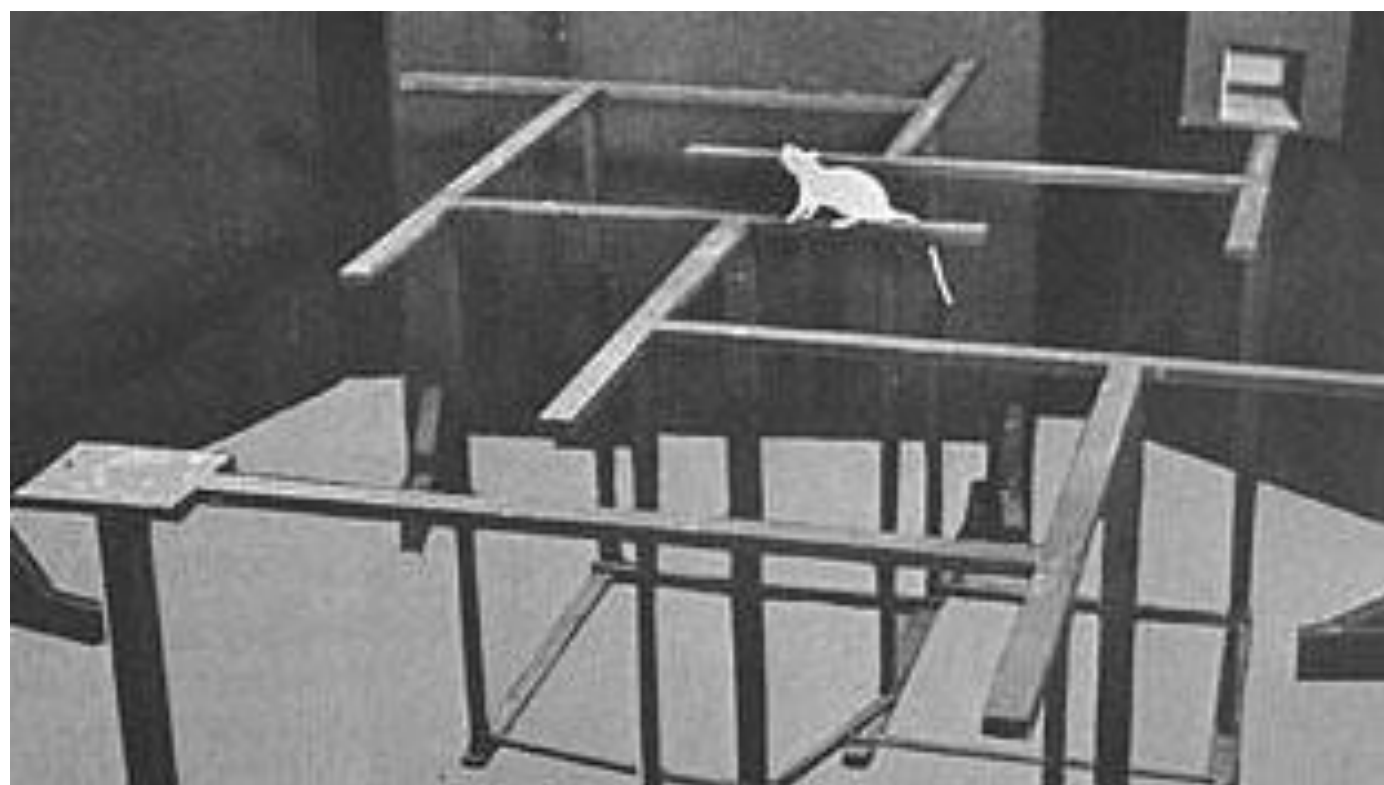
Tolman's Cognitive Behaviorism

- Tolman disagreed with Hull and believed that it would be more useful to analyze behavior on a broader level.
- Behavior is more than just a chain of discrete responses attached to discrete stimuli.

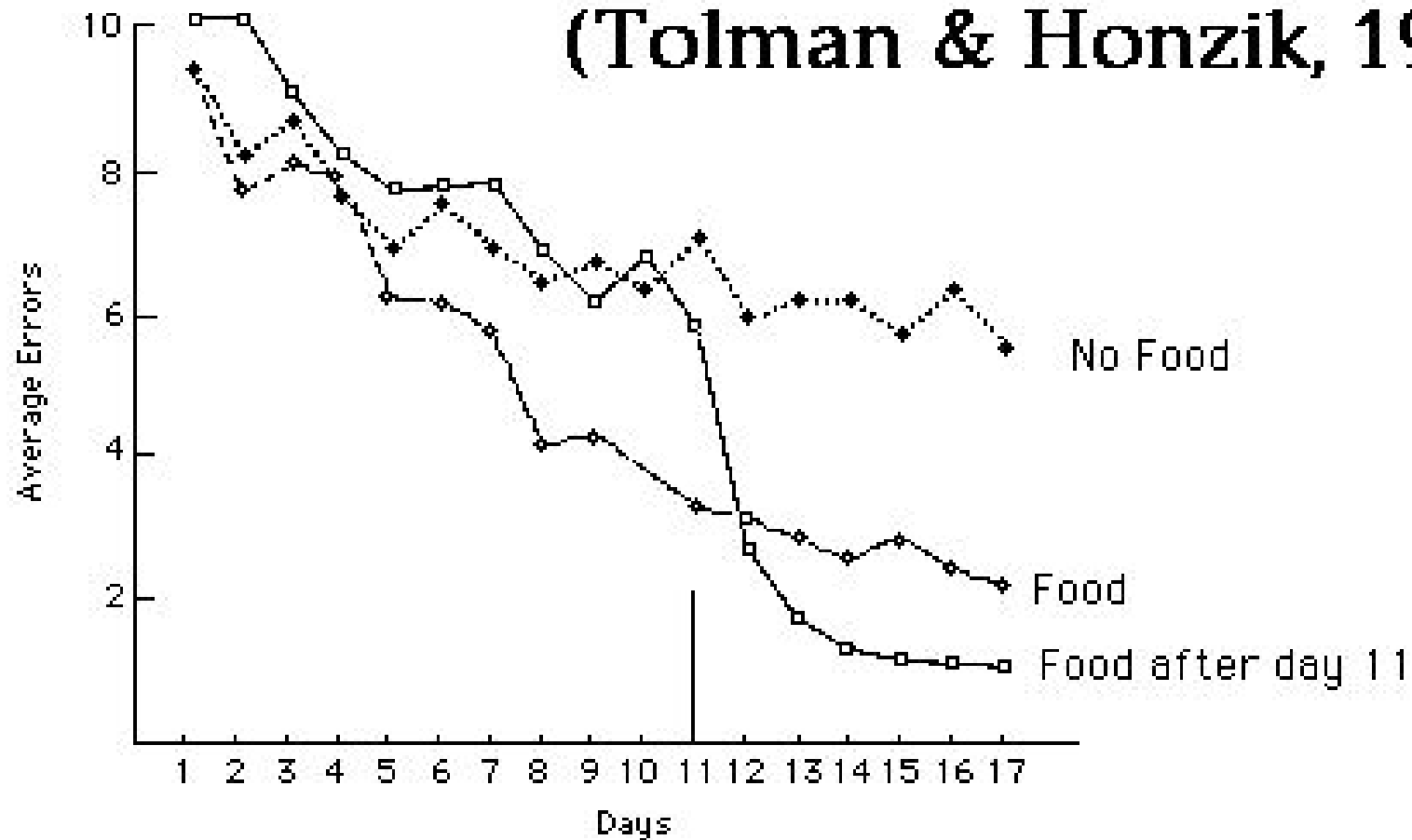
Tolman's Cognitive Behaviorism

- Tolman's *cognitive behaviorism* utilizes intervening variables, usually in the form of hypothesized cognitive processes, to help explain behavior.
- We create *cognitive maps* which are mental representations of our spatial surroundings.





Latent Learning (Tolman & Honzik, 1930)

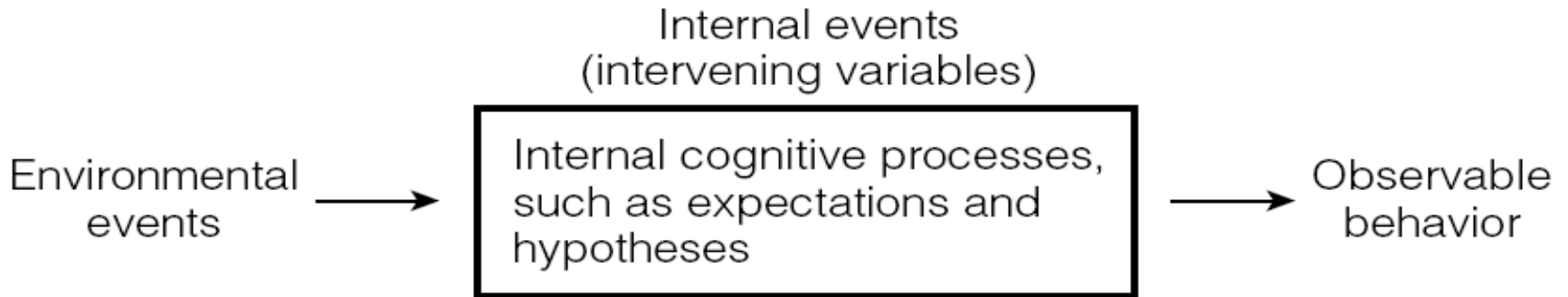


learning vs. performance

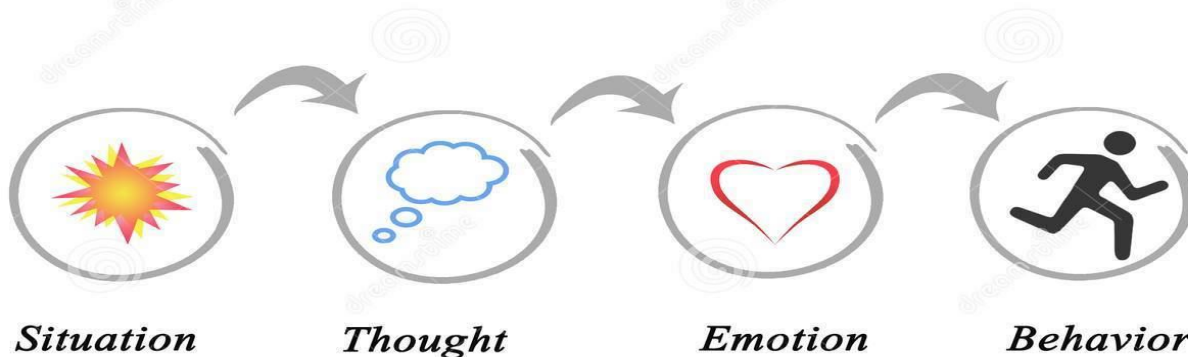


Cognitive Behaviorism

FIGURE 1.3 In Tolman's cognitive behaviorism, theorists make use of intervening variables, in the form of hypothesized cognitive processes, to help explain the relationship between environment and behavior.



Cognitive Model



Vacuum technology

← → ↺ 🏠 🔒 https://www.amazon.com/iRobot-Bundle-Connected-Mapping-Edge-Sweeping/dp/B07NVSPRYD/ref=asc_df_B07NVSPRYD/?tag=hyprod-20&linkCode=... 📄 📌 🌱 effy

Amazon Home

Shop by Room

Discover

Shop by Style

Home Décor

Furniture

Kitchen & Dining

Bed & Bath



Your medication, delivered



Learn more >

Home & Kitchen > Vacuums & Floor Care > Vacuums > Robotic Vacuums



Roll over image to zoom in



iRobot Roomba i7 Robotic Vacuum Bundle- Alexa Connected, Home Mapping, Great for Pet Hair (+1 Extra Edge-Sweeping Brush)

by iRobot

★★★★★ 45 ratings | 31 answered questions

Price: **\$699.00** & **FREE Shipping**. [Details](#)

[Free Amazon product support included](#)

- Power-Lifting Suction provides 10X the air power* for improved pick-up performance. Great for households with pets. Premium 3-Stage Cleaning System attacks the dirt you see and the allergens don't. *(Compared to Roomba 600 Series and AeroVac System)
- Patented iAdapt 3.0 Navigation with vSLAM technology allows Roomba to easily navigate and efficiently clean an entire level of your home.
- Imprint Smart Mapping enables Roomba to learn and map your home, letting you control which rooms are cleaned and when.
- High-Efficiency Filter captures 99% of allergens from cats and dogs - Dual Multi-Surface Rubber Brush adjust and flex to keep in constant contact with your carpets and hard floors.
- Box Content: 1 Roomba i7 Robot Vacuum, 1 Home Base Charging Station, 1 Cord, 1 Dual Mode Virtual Wall Barrier, 2 Extra High-Efficiency Filters, 2 Extra Side Brushes

[Compare with similar items](#)

New (2) from **\$699.00** + **FREE Shipping**

[Report incorrect product information.](#)



iRobot Roomba E5 (5150) Robot Vacuum - Wi-Fi Connected, Works with Alexa, Ideal for Pet Hair, Carpets, Hard, Self-Charging

★★★★★ 586

\$329.00 **prime**

[Ad feedback](#)



Self-Driving Cars



Bandura's Social Learning Theory

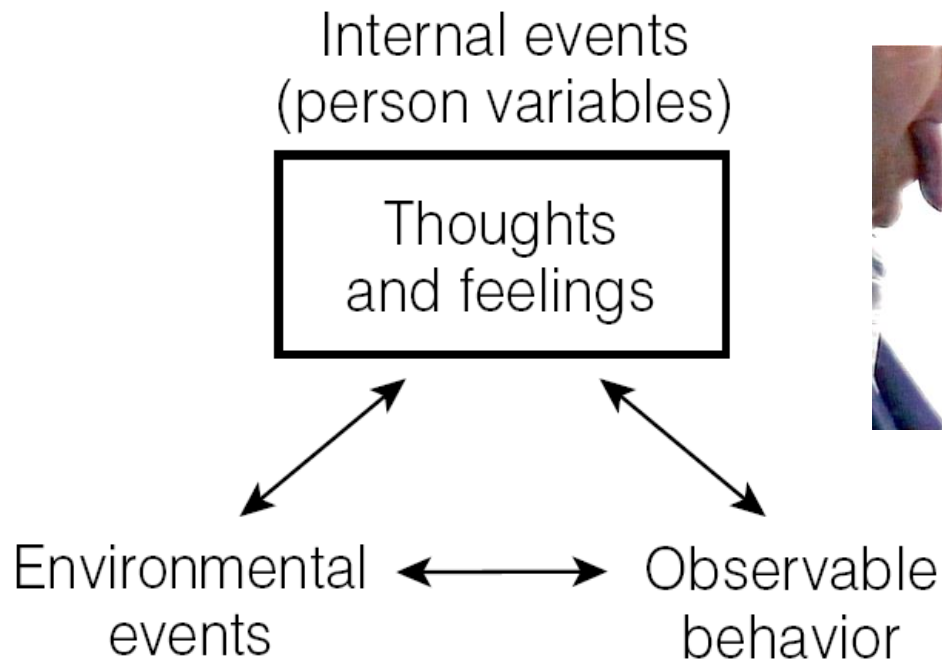
- A cognitive-behavioral approach that strongly emphasizes the importance of observational learning and cognitive variables in explaining human behavior



Social Learning Theory

FIGURE 1.6

Bandura's model of reciprocal determinism, in which observable behavior, environmental events, and internal events are all viewed as interacting with each other.



Why not study internal events?

- Our assessments of internal thoughts and feelings thus are often unreliable.
 - Example: the task of teaching a young boy to correctly label the feeling of pain.
- It is often difficult to determine the actual relationship of thoughts and feelings to behavior.
 - Example: the act of providing help in an emergency.
- We do not have any means of directly changing these internal events.
- Such explanations are sometimes only pseudo explanations.
 - Example, I “feel like going to the movies.”

How can we predict popularity?

- How do you predict which food will be popular or which toy will be the “it” toy this year?



Traditional Methods

- Standard marketing techniques
 - Focus groups
 - Questionnaires
 - Simulated choice tests
 - Market tests.
 - Mechanical TURK

Mechanical Turk is a marketplace for work.

We give businesses and developers access to an on-demand, scalable workforce. Workers select from thousands of tasks and work whenever it's convenient.

743,363 HITS available. [View them now.](#)

Make Money by working on HITs

HITs - *Human Intelligence Tasks* - are individual tasks that you work on. [Find HITs now.](#)

As a Mechanical Turk Worker you:

- Can work from home
- Choose your own work hours
- Get paid for doing good work



[or learn more about being a Worker](#)

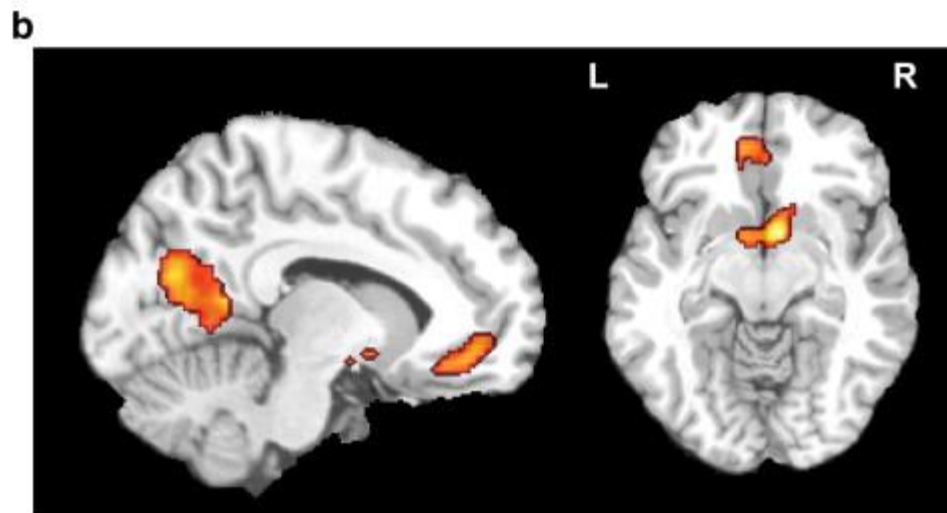
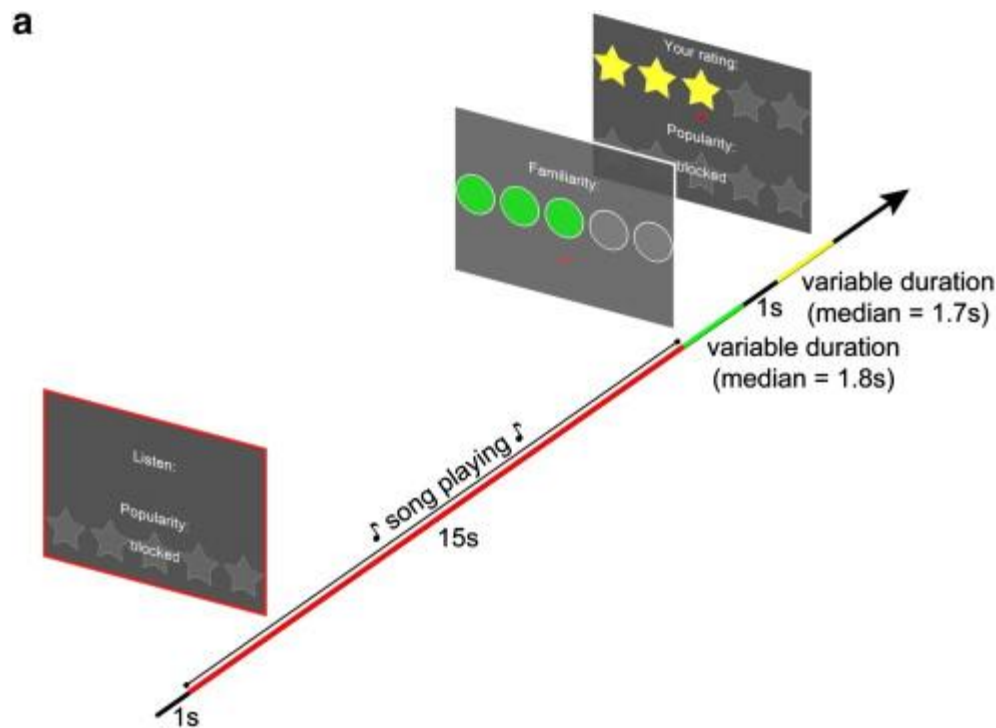
Get Results from Mechanical Turk Workers

Ask workers to complete HITs - *Human Intelligence Tasks* - and get results using Mechanical Turk. [Get Started.](#)

As a Mechanical Turk Requester you:

- Have access to a global, on-demand, 24 x 7 workforce
- Get thousands of HITs completed in minutes
- Pay only when you're satisfied with the results

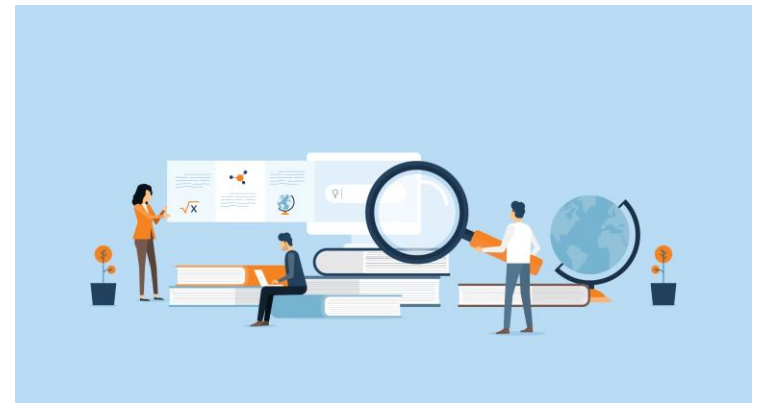




Gregory S. Berns, Sara E. Moore. **A neural predictor of cultural popularity.** *Journal of Consumer Psychology*, 2011

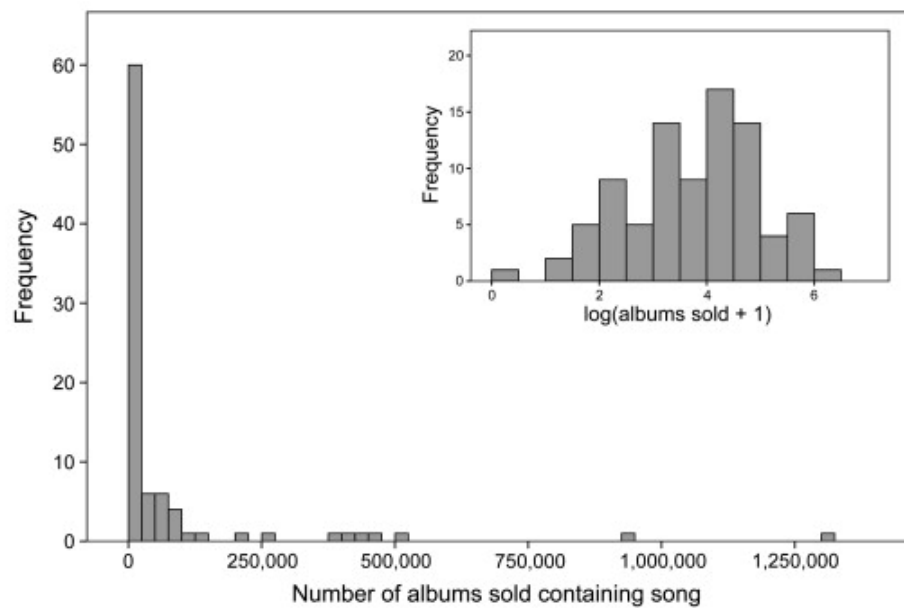
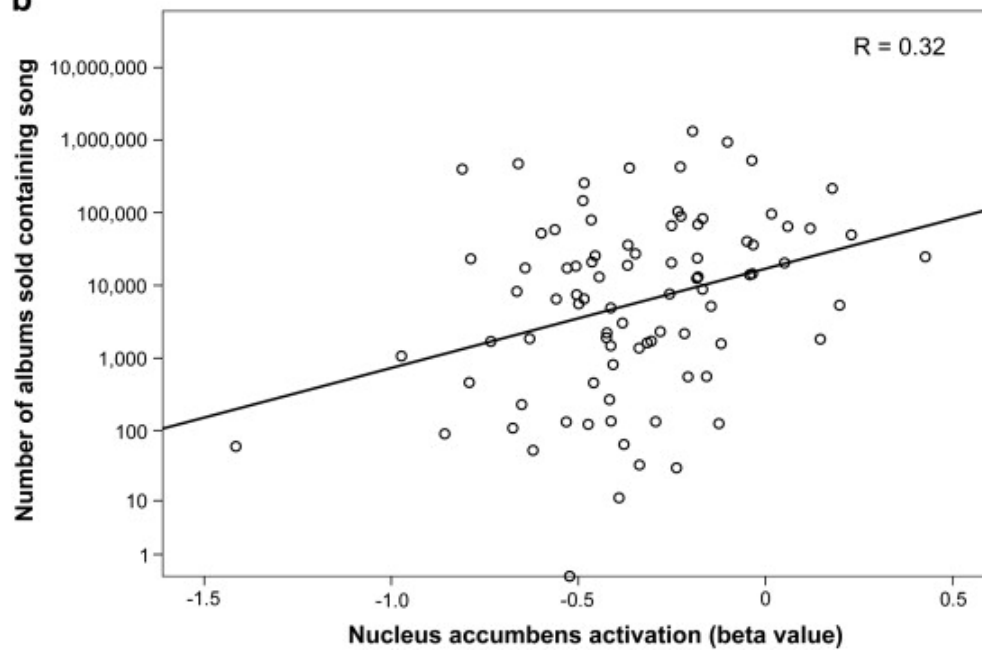
Findings

- Orbitofrontal cortex and ventral striatum, is predictive of future purchasing decisions
- used fMRI to measure the brain responses of a relatively small group of adolescents while listening to songs of largely unknown artists.

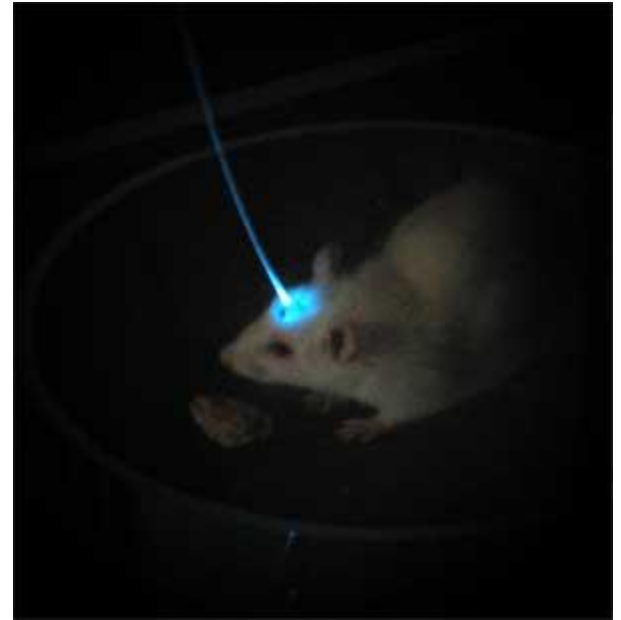
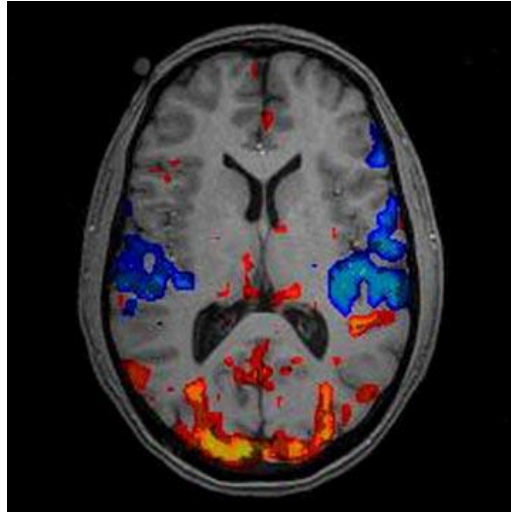




- Although subjective likability of the songs was not predictive of sales, activity within the ventral striatum was significantly correlated with the number of units sold.

a**b**

Today's Neuroscience



Presented clip



Clip reconstructed
from brain activity



Jack Gallant. Current Biology Sept 22, 2013

Skinner's Molar View

- Only reflexive behaviors are automatically elicited by the stimuli that precede them.
 - Example: salivating in response to food
- Reflexive behaviors are distinguished from behaviors that are controlled by their consequences, which are more flexible and less predictable.
 - Example: rat is running through the maze because such behavior has *in the past* resulted in food.
- We should explain the behavior by referring to past experience.

Skinner's Countercontrol

- Understand how environment affects us
- Change environment to suite our desired behavior



Skinner's View of Genetic Factors

- Behavior is fundamentally the result of the interaction between genes and the environment.
- However, genetic factors are largely unmodifiable.
- To assume that a behavior pattern has a strong genetic basis is to assume also that little can be done to alter it.
- Skinner believed that genes have a great impact on behavior, but behaviors can change.
- Behaviors that lead to favorable outcomes are more likely to be repeated, whereas those that do not lead to favorable outcomes are less likely to be repeated.

Skinner & Free Will

- Not a measurable quantity and thus should not be in empirical research
 - Unpredictable aspects are from lack of data influencing the behavior
 - Aka an error variance
- Behavior can be influenced in the most subtle and undetectable ways
 - Lighting (Hawes et al. ,2012; Slegers et al., 2013; Yang & Jeon, 2020)
 - Wall color (Ainsworth et al., 1993; Read et al., 1999; Al-Ayash et al., 2016; Sakuragawa et al., 2005 [wall material matters])

Skinner & Free Will

- Perception of free will broken when influencers are aversive
 - Recommends utilizing positive reinforcement to change behavior
 - Society/social “engineering”



Behavior Analysis

- Grew out of radical behaviorism.
- Applied Behavior Analysis is a technology of behavior in which basic principles of behavior are applied to real-world issues.
 - also sometimes referred to as *behavior modification*
 - Example: helping people with clinical disorders, such as phobias and schizophrenia, autism...



https://www.baylor.edu/soe/edp/index.php?id=935205

170%



applied behavior analysis and baylor



Learn to help children with
developmental disabilities
through a

Baylor
ABA
graduate program

CLICK to join the
MAILING LIST

Behavioral economics

- **Behavioral economics**, along with the related sub-field **behavioral** finance, studies the effects of psychological, social, cognitive, and emotional factors on the **economic** decisions of individuals and institutions and the consequences for market prices, returns, and resource allocation.

COVID-19 VACCINATION INCENTIVE PROGRAM



METROPOLITAN
COMMUNITY COLLEGE





MENTALISM

**WATSON'S
BEHAVIORISM**

Tolman

Skinner

Social Learning CBT

Summary of Historical Significance

- Aristotle assumed that knowledge is largely gained from experience
- Descartes proposed that involuntary behaviors are automatically elicited by external stimulation
- British empiricists argued that all knowledge is a function of experience
- Structuralists hold that we need to identify the basic elements that compose it
- Darwin's theory of evolution says that adaptive characteristics evolve through the process of natural selection.
- Functionalists study publicly observable behavior and the environmental events that influence them

Summary of Schools of Behaviorism

- Watson's methodological behaviorism
- Hull's neobehaviorism
- Tolman's cognitive behaviorism
- Bandura's social learning theory
- Skinner's radical behaviorism

This text will introduce us to

- Classical Conditioning, in which reflexive behaviors come to be elicited in new situations
- Operant Conditioning, in which the probability of a behavior is influenced by its consequences